Matrices

Mind map

Definition and types
- Order $m \times n$; a_{ij}
- Diagonal, scalar, triangular
- Singular iff $|A| = 0$

Equality and addition
- Same order, equal entries
- $\operatorname{tr}(A \pm B)$ adds
- $|A + B| \neq |A| + |B|$

Scalar multiple
- $kA = [ka_{ij}]$
- $|kA| = k^n|A|$

Multiplication
- $(m \times n)(n \times p)$
- Row $\times$ column
- $|AB| = |A||B|$; $AB \neq BA$

Commuting and identities
- $AB = BA$ for $(A \pm B)^2$
- $AB = -BA$ anti-commute
- $(A + kI)^n$ binomial

Powers and characteristic equation
- $A^2 - (\operatorname{tr} A)A + |A|I = O$
- Reduce powers / divide
- Pattern, $\lambda I + N$

Special matrices
- $A^2 = A$: $(I + A)^n = I + (2^n - 1)A$
- $A^2 = I$: $2^{n-1}(I + A)$
- Nilpotent: finite expansion

Transpose
- $(AB)^T = B^T A^T$
- $|A^T| = |A|$

Symmetric and skew
- $A^T = \pm A$
- Skew: diagonal 0, odd order $|A| = 0$
- $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

Orthogonal
- $AA^T = I$, $A^{-1} = A^T$
- Rows unit and perpendicular
- $R(\theta)^n = R(n\theta)$

Adjoint
- $A \operatorname{adj} A = |A|I$
- $|\operatorname{adj} A| = |A|^{n-1}$
- $\operatorname{adj}(\operatorname{adj} A) = |A|^{n-2}A$

Inverse
- $A^{-1} = \operatorname{adj} A/|A|$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $AB = O$: both singular

Elementary operations
- Rows: $[A|I] \to [I|A^{-1}]$
- Columns only for columns

System and rank
- $AX = B$, $X = A^{-1}B$
- $(\operatorname{adj} A)B$ test if $|A| = 0$
- Rank = largest non-singular minor

Permutation and quadratic form
- PA rows, AP columns
- $P^k = I$, lcm of cycles
- $X^T AX = \sum a_{ij}x_i x_j$

Counting and patterns
- $\operatorname{tr} A^T A = \sum a_{ij}^2$
- Symmetric: $s^{n(n+1)/2}$
- Use $|A| \neq 0$ complement